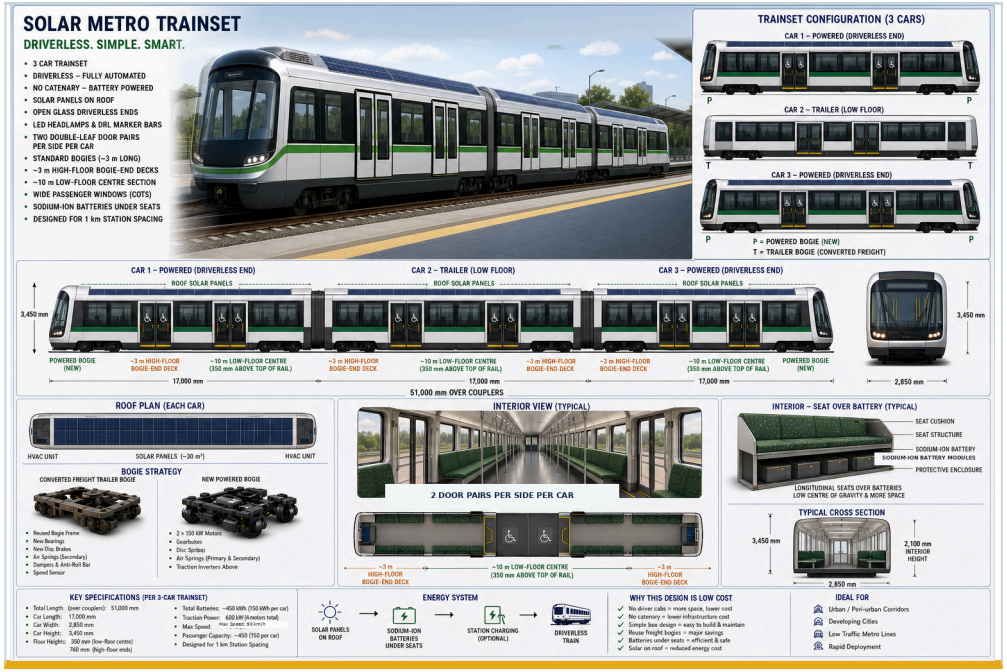
FOR CITIES THAT WANT USEFUL RAIL, BUILT TO LAST

Give your city a railway it can own.

OpenSourceRail helps public transport leaders plan, build and operate modern urban rail while keeping the skills, jobs and decisions that matter close to home.

Start with a clear service plan and a maintainable asset base. Test the network before construction, connect every design change to its standards evidence, and grow from a depot or pilot corridor toward dependable citywide service. The public portfolio contains 265 developing-world city models across 43 country programmes, plus one European comparison model.



Better access

Plan lines, stations, service and capacity around how people move through the city.

Local value

~79% modelled domestic value across the public portfolio.

Lower risk

Simulate designs and identify affected standards before committing to hardware and works.

266 models

265 developing-world city packages plus one European comparison model.

WHAT THE CITY GETS

One programme from first idea to everyday service.

FOR PASSENGERS

More legible routes, useful service plans and a railway designed around real stations, capacity and daily journeys.

FOR LOCAL INDUSTRY

Open interfaces and repeatable packages support local fabrication, integration, training and long-term maintenance.

FOR DECISION-MAKERS

See cost, energy, delivery sequence and evidence together before a city commits to a build path.

FROM AMBITION TO OPERATION

01 Choose a city, corridor and service ambition.

02 Generate and compare a deterministic city digital twin.

03 Review routes, stations, fleet, energy, cost and buildability.

04 Turn the selected design into CAD, IFC, quantities and delivery packages.

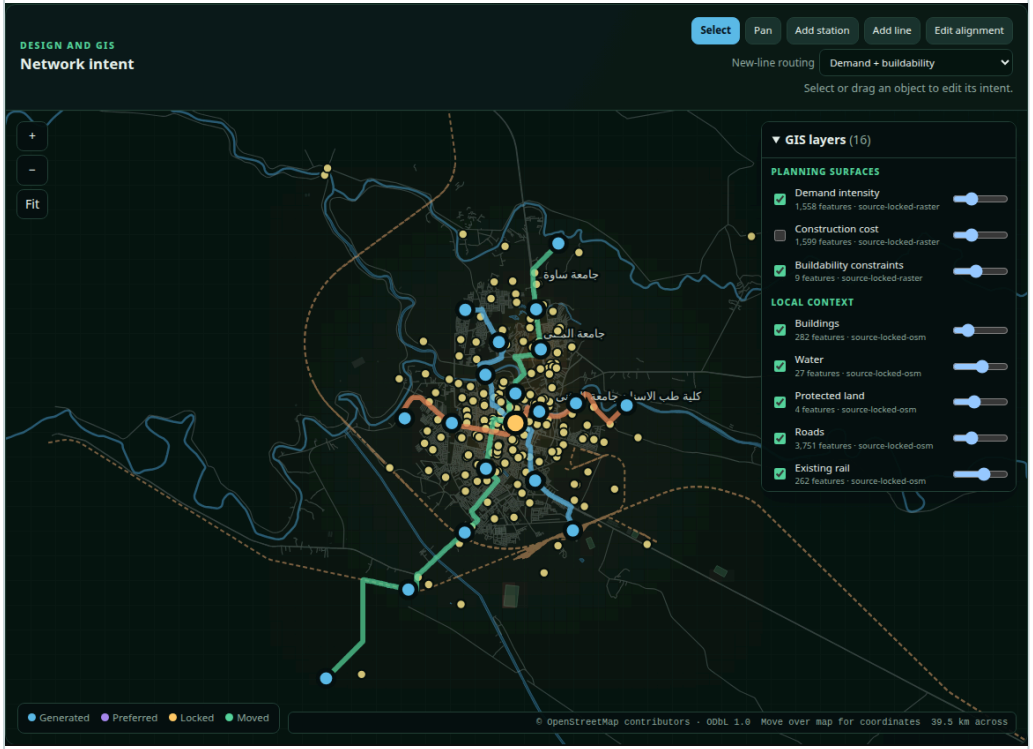
05 Start with a depot, workshop or pilot corridor and train local teams.

06 Expand only as field evidence, assurance and operating capability mature.

A CITY OPERATING ADVANTAGE

Make standards adherence a living digital thread.

One connected workbench gives city teams a shared view of the network, projects, assets, work orders, simulations and evidence, with every approval authority still explicit.



CHANGE ONCE; KNOW WHAT MUST BE REASSESSED

The standards registry links 23 standards and legal records to 16 control objectives and 15 deterministic repository checks. Evidence is hashed, versioned and traced across civil, rolling stock, electrical, software, cyber, BIM, quality, operations and maintenance.

SHAPE THE SERVICE

Explore corridors, stations, demand, capacity, water crossings and civil choices over local GIS before a route becomes a construction commitment.

CONTROL DELIVERY

Turn an approved design into assets, work packages, schedules, order-by dates, local/imported cashflow and manufacturing quality gates.

ASSESS THE IMPACT

Compare two generated baselines to identify affected checks, controls, standards and failure modes before accepting a design change.

BUILD CREDIBLE EVIDENCE

Keep GIS, CAD, IFC, cost, simulation and test outputs tied to source and tool hashes. Machines check coherence; competent people and authorities decide conformity.

ONE EVIDENCE PATH

A requirement leads to a control, design artefact, FMEA item, test, evidence record, reviewer and release gate. Missing deployment inputs, field tests, competent review or current licensed clauses remain visible as open obligations.

Scope	Trace	Verify	Approve
rules and profile	controls and risks	tests and hashes	human authority

DETERMINISTIC BY DESIGN

DIGITAL THREAD	REPOSITORY STATUS
Standards / legal records	**23 versioned records**
Control objectives	**16 mapped controls**
Automated checks	**15 hashed checks**
Change-impact index	**Generated per baseline**
Physical release	Blocked pending real evidence

A RESPONSIBLE PATH TO SERVICE

The digital thread reduces omissions, redesign and approval preparation time. It is planning and engineering-screening evidence—not a safety certificate, construction release, supplier quote, approval or operating authority. Licensed standards, deployment evidence, independent assessment and authority acceptance remain mandatory.

github.com/modernecotech/OpenSourceRail